

Titouan Pottier

Seeking a CIFRE PhD position in Machine Learning
MVA – Mathematics, Vision, Learning

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in Titouan Pottier

EDUCATION

- **Master's degree, MVA - ENS Paris-Saclay** 2025-2026
MVA (Mathématiques, Vision, Apprentissage)
 - Deep Learning, Computer vision, Computational Statistics, Generative Models for Images, Reinforcement Learning, Representation Learning, Multimodal Explainable Artificial Intelligence (XAI), Distributed Training of Large-Scale Models.
- **Engineering Student - École des Ponts ParisTech** 2022-2025
Mathematical and Computer Engineering Department
 - Image Processing and Computer Vision, Machine Learning, Deep Learning, Convex Optimization, High-Dimensional Statistics, Stochastic Processes, and Applications.
- **Pierre Corneille High School, Rouen** 2020-2022
Intensive College Preparatory Program for Elite Universities Écoles MPSI-PSI*
 - Two years of intensive preparation in mathematics and physics for the national engineering school entrance exam

EXPERIENCES

- **Research Intern in Computer Vision (Incoming)** April-September 2026
Multiframe Super-Resolution with Differentiable Optics Enhance Lab, Paris
 - Supervised by Prof. Jean Ponce
 - Extension of multiframe super-resolution pipelines (Lucas–Kanade Reloaded) with a differentiable optical forward model
 - Joint estimation of optical aberrations and image reconstruction from raw image bursts
 - Modeling wavelength-dependent PSFs and chromatic aberrations using physically interpretable representations (Zernike polynomials)
 - Study of identifiability, robustness, and reconstruction quality under realistic image formation models
 - Experiments on synthetic and real raw burst datasets with controlled optical aberrations
- **Research Intern in Deep Learning for Bioinformatics** February-July 2025
Antigen-antibody affinity prediction Scuola Normale Superiore, Pisa
 - Prediction from amino acid sequences using Protein Language Models
 - Usage of deep learning methods to improve embeddings
- **Intern Deep Learning Engineer in 3D Medical Imaging** July-December 2024
LLMs Exploration Internship in 3D Medical Imaging General Electric Healthcare, Paris
 - Implementation of Advanced RAG for internal code generation
 - Driving internal algorithms via LLM
 - Usage of Vision Transformers models for CT scan volumes analysis
- **Drug Design by Sanofi** January-June 2024
Drug design using generative AI Sanofi, Paris
 - Use broadcast algorithms for Molecular Docking (DiffDock)
 - application to the KRas protein

INTERESTS

- **Student Office - BDE - École des Ponts ParisTech**
- **Competitive Sports** 2010-Now
 - Member of the Football Team at École des Ponts et Chaussées
 - Competitive Athletics: Triathlon and 400m at the national level

ADDITIONAL SKILLS

Programming Languages: Python, C++, Julia, Castem, R, Pack Office

French: Native

English: B2 - European Class from 6th to 12th grade, Fluent-TOEIC Listening and Reading (785)

German: B1

International Openness : 1-Week exchange at İTÜ (Istanbul Technical University) in 2023 | 2-Week school exchange in Germany in 2017